

Lecture 8 - Social Neuroscience: Understanding others

The Social Brain: Critical Perspectives on Science, Society and Neurodiversity

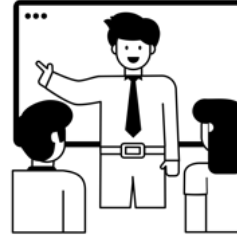
Richard Ramsey



Today

Part 1

- Social Neuroscience:
Understanding others



Part 2

- Read articles and discuss



Understanding others

For an overview, See Chapter 6 from Ward ([2022](#))



Overview

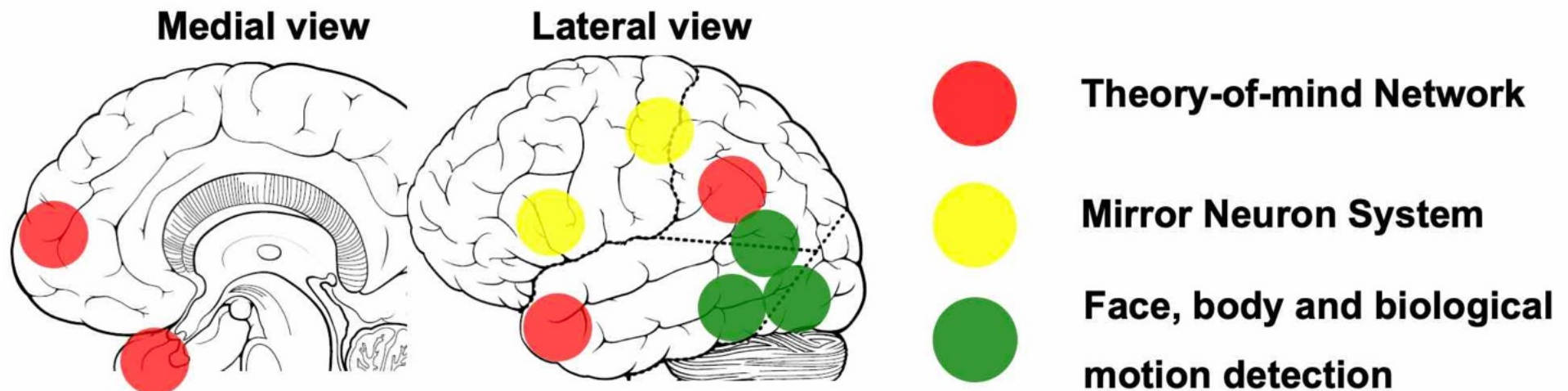
- Recap “Social Brain” circuits
- Background
- Cognitive and brain systems
 - Mirror neuron system
 - Theory of mind network



Recap social brain circuits



Social brain circuits



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(Adolphs 2009, Stanley & Adolphs 2013)

Background



Mindreading?



Beliefs about beliefs

first-order



second-order



third-order



Spontaneous



(Frischen et al. 2007)

Why is it important?



prediction

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Why is it important?



manipulation



Why is it important?



joint action

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Cognitive and brain systems

- Mirror neuron system
- Theory of mind network



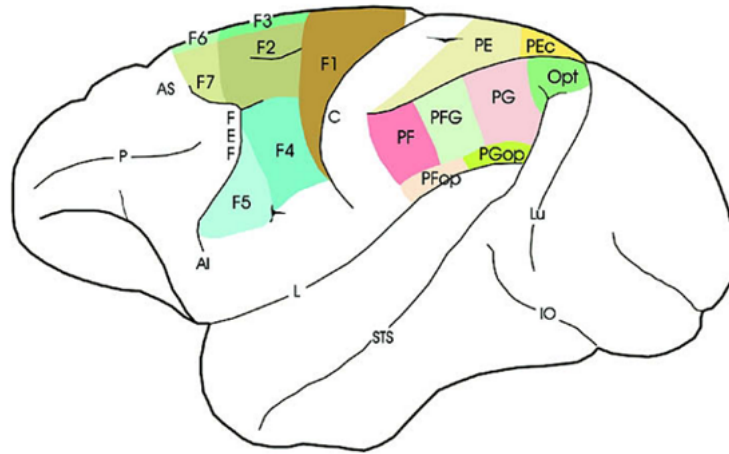
Mirror neurons



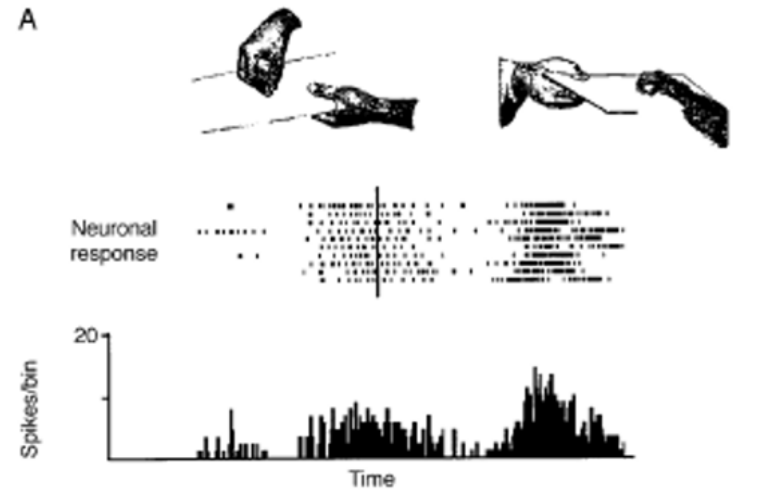
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Mirror neurons in macaques



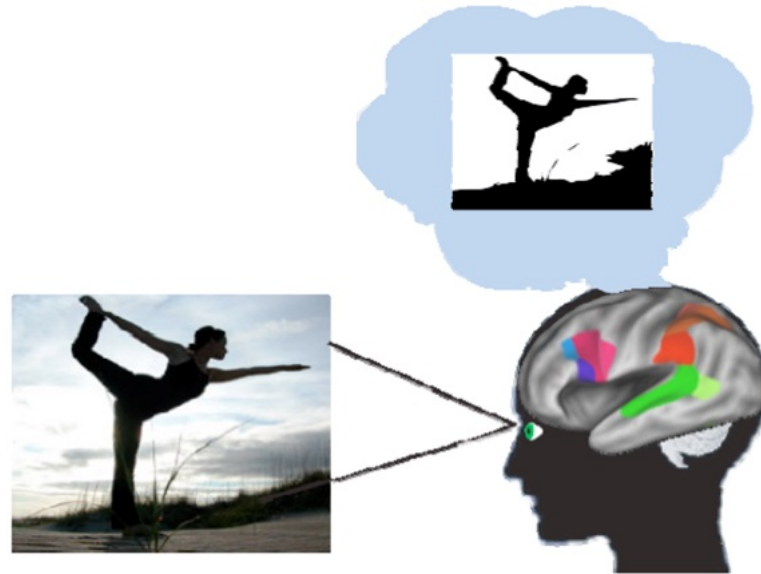
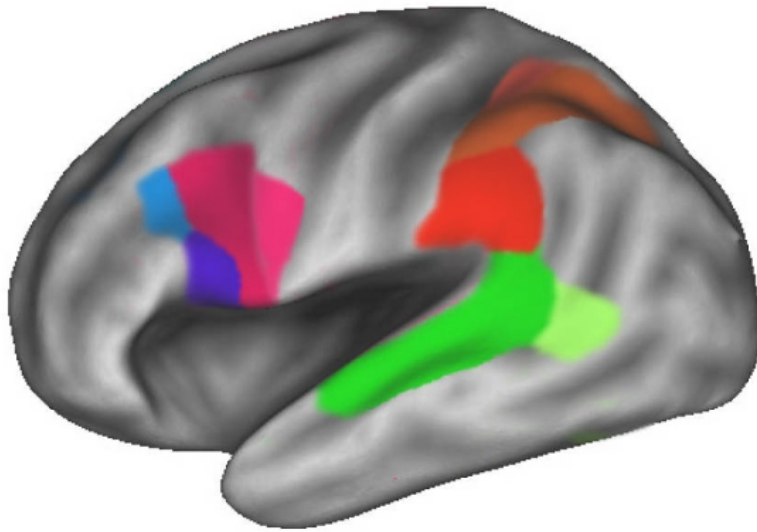
F5 and PF



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(di Pellegrino et al 1992, Fogassi et al 2005, Gallese et al 1996)

Mirror neuron system in humans



The direct matching hypothesis



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(Gazzola & Keysers 2009, Rizzolatti et al 2001)

Theory of mind

ToM = reasoning about mental states



Le Tricheur à l'as de carreau by Georges de la Tour (c. 1635)



Theory of mind development

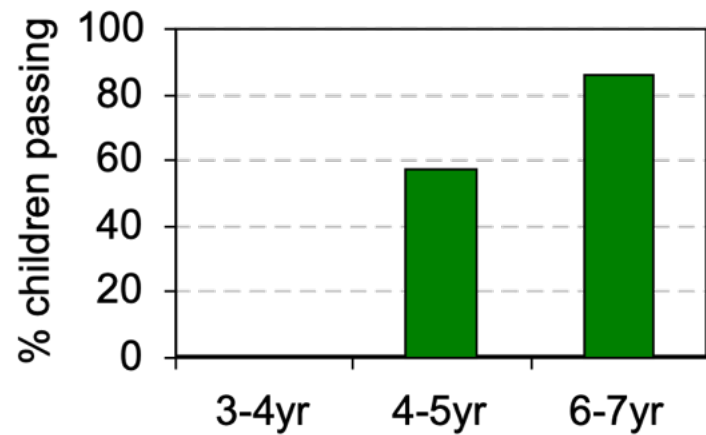


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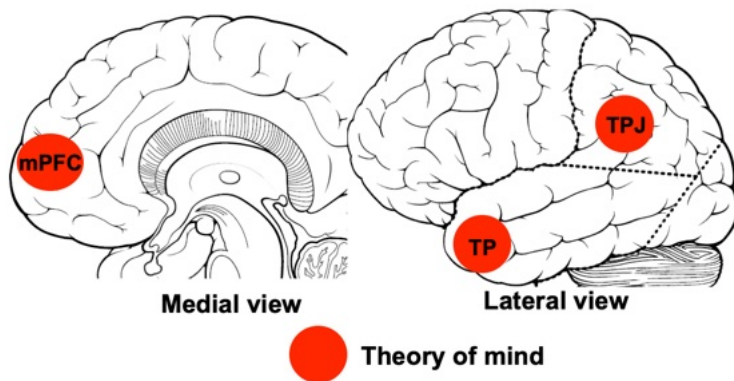
The false belief task



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(Wimmer & Perner 1983)

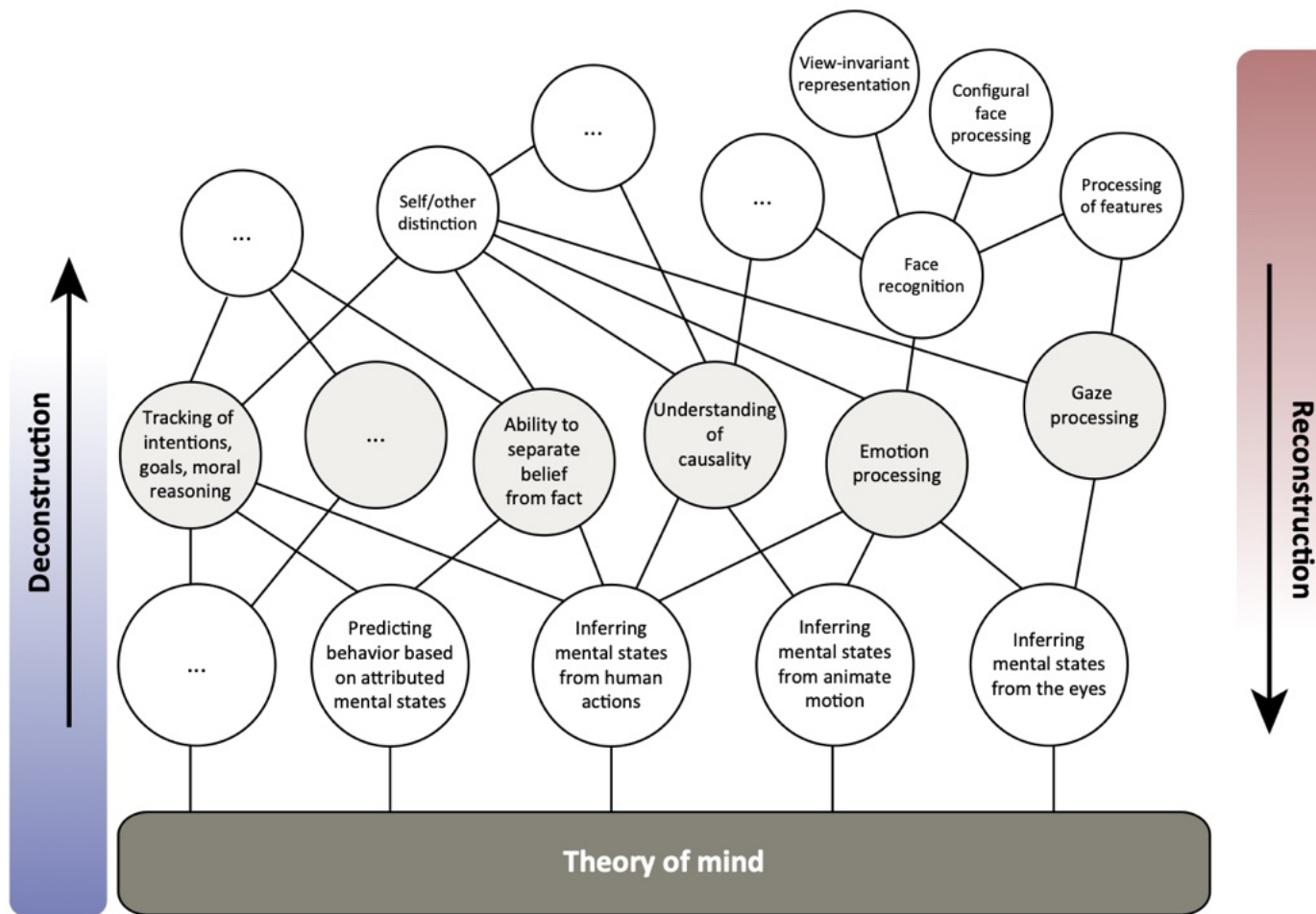
Brain circuits



- mPFC: person knowledge
- TPJ: false belief, perspective taking
- TP: social scripts, emotion, convergence ?

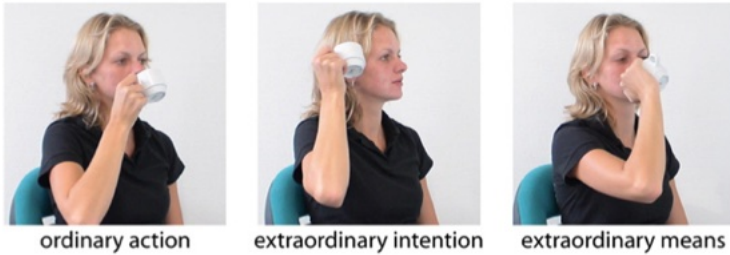


Cautions

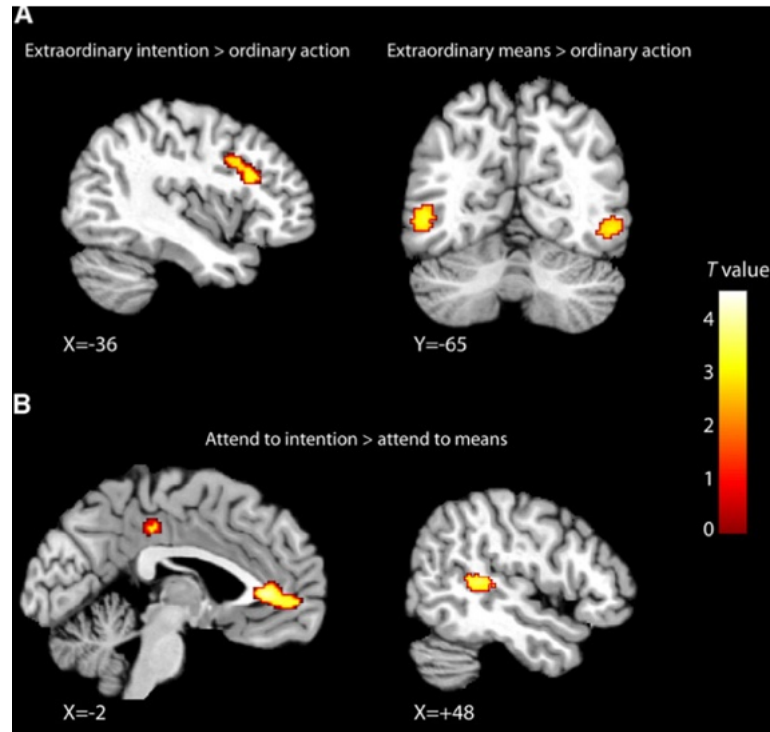
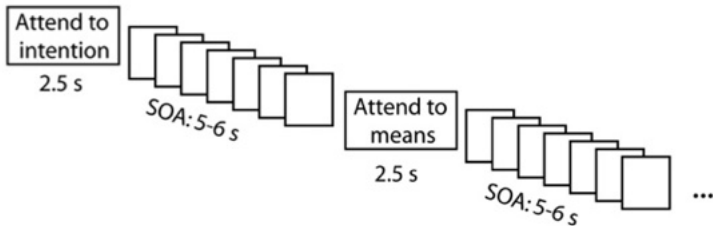


Mirroring and theory of mind

A



B



Mirroring and theory of mind

- Complementary roles in social cognition.
- How do they influence each other?
- Lots of open and unanswered questions.
- Essay questions?



Take a break



Part 2 - Read and discuss



Discussion material

- break into small groups (~ 5 per group)
- discuss aspects of the lecture
- discuss aspects of the journal article: <https://www.sciencedirect.com/science/article/pii/S1364661314002502?via%3Dihub>



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